



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
Student Name	Joshi Rushitha Kasturi
Roll Number	2520030204
Branch	CSE
Course Outcome	AVL

Case Study Topic: JobMatch – Intelligent Job Recruitment & Talent Matching System

Work Done Summary:

In this case study, we implemented an **AVL Tree** for the **JobMatch – Intelligent Job Recruitment & Talent Matching System**. The AVL Tree stores candidate records in a balanced manner, enabling efficient insertion, searching, and retrieval operations. As new candidates register, their information is inserted into the AVL Tree, and balancing operations are performed automatically using LL, RR, LR, and RL rotations whenever necessary. This ensures that the height of the tree remains balanced and all operations execute efficiently in $O(\log n)$ time.

Implementation Details

1. Created a Candidate Management module to store candidate details.
2. Implemented AVL Tree insertion for candidate records.
3. Maintained balance using AVL rotations (LL, RR, LR, RL).
4. Implemented candidate search functionality.
5. Displayed candidate records using tree traversal.
6. Compared AVL Tree performance with a normal BST.
7. Verified that AVL Tree operations maintain $O(\log n)$ complexity.

Result: Screenshot/Output:

```
PS C:\Users\Joshi_Rushitha\OneDrive\Desktop\JobMatchProject> java Main
JOBMATCH RECRUITMENT SYSTEM

BST Traversal
101 Rahul Java 3 years
102 Sneha Python 2 years

Project Executed Successfully
PS C:\Users\Joshi_Rushitha\OneDrive\Desktop\JobMatchProject>
```

Git Hub Link:

<https://github.com/joshirushithakasturi-arch/DSAA2>

Student Signature	Faculty Signature



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Faculty Name	Mohammed Razia Alangir Banu
Student Name	Joshi Rushitha Kasturi
Roll Number	2520030204
Branch	CSE
Course Outcome	CO-4 (Dijkstra's Algorithm)

Case Study Topic: **JobMatch – Intelligent Job Recruitment & Talent Matching System**

Work Done Summary:

In this case study, we implemented **Dijkstra's Algorithm** for the **JobMatch – Intelligent Job Recruitment & Talent Matching System**. The algorithm is used to find the shortest path between different recruitment nodes such as candidates, jobs, recruiters, and interview locations. By calculating the minimum cost path, the system can optimize recruitment processes and improve decision-making. The implementation efficiently determines the shortest route from a source node to all other nodes in the recruitment network.

Implementation Details

- Created a weighted graph representing recruitment connections.
- Assigned weights to edges based on recruitment cost or distance.
- Implemented Dijkstra's Algorithm using adjacency matrices.
- Selected a source node for shortest path computation.
- Calculated minimum distances from the source to all nodes.
- Updated distances whenever a shorter path was found.
- Displayed shortest paths for recruitment analysis.

Result: Screenshot/Output:

```
JOBMATCH RECRUITMENT SYSTEM
Shortest Paths from Source Node 0
To Node 0 = 0
To Node 1 = 4
To Node 2 = 6
To Node 3 = 9
To Node 4 = 11
Recruitment Route Analysis Completed
PROJECT COMPLETED
```

Git Hub Link:

<https://github.com/joshirushithakasturi-arch/DSAa2>

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Student Name	Joshi Rushitha Kasturi
Roll Number	2520030204
Branch	CSE
Course Outcome	C0-3 (BFS DFS)

Case Study Topic: JobMatch – Intelligent Job Recruitment & Talent Matching System

Work Done Summary:

In this case study, we implemented **BFS (Breadth First Search)** and **DFS (Depth First Search)** algorithms for the **JobMatch – Intelligent Job Recruitment & Talent Matching System**. These graph traversal algorithms are used to analyze recruitment networks and candidate-job relationships efficiently. BFS explores candidate connections level by level, while DFS explores deeper relationships before backtracking. The implementation helps in identifying candidate networks, skill dependencies, and recruitment pathways within the system.

Implementation Details

1. Created a graph structure to represent candidate-job relationships.
2. Implemented BFS traversal using a queue data structure.
3. Implemented DFS traversal using recursion.
4. Connected candidate and job nodes through graph edges.
5. Traversed the recruitment network using BFS.
6. Traversed the recruitment network using DFS.
7. Compared BFS and DFS traversal results.

Result: Screenshot/Output:

```
PS C:\Users\Joshi_Rushitha\OneDrive\Desktop\JobMatchProject> java Main
JOBMATCH RECRUITMENT SYSTEM

BST Traversal
101 Rahul Java 3 years
102 Sneha Python 2 years

Project Executed Successfully
PS C:\Users\Joshi_Rushitha\OneDrive\Desktop\JobMatchProject>
```

Git Hub Link:

<https://github.com/joshirushithakasturi-arch/DSAa2>



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